The Meixner–Pollaczek Matching Matrix: Approximating Meixner–Pollaczek Functions by Meixner–Pollaczek Polynomials

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Abstract

The Meixner–Pollaczek polynomials $P_n^{(1/4)}(x)$ (at $\phi = \pi/2$) are orthogonal with respect to the weight $W(x) = |\Gamma(1/4 + ix)|^2$ on $\mathbb{R}$. Absorbing the square root of the weight into the polynomial yields the *Meixner–Pollaczek functions* $\Phi_n = P_n\sqrt{W}/\sqrt{h_n}$, orthonormal in $L^2(\mathbb{R})$; by a theorem of Bump and Ng these are precisely the Mellin transforms of the Hermite functions restricted to the critical line. We compute the matching matrix b_{mk} expressing the $L^2(W)$-projection of Φ_m onto the polynomial basis, the exact analogue of the Hermite matching matrix. The sign pattern and parity structure of the Hermite case survive, but its two salient features do not: the entries have no apparent algebraic structure, and the coefficients decay only algebraically, $b_{0k} \sim k^{-\alpha}$ with $\alpha \approx 3/4$, rather than geometrically. The obstruction is a branch point of $\sqrt{W}$ at distance $1/4$ from the real axis.

1 Setup

The Meixner–Pollaczek polynomials with $\lambda = 1/4$, $\phi = \pi/2$ satisfy

$$(n+1)\,P_{n+1}(x) = 2x\,P_n(x) - \big(n - \tfrac{1}{2}\big)P_{n-1}(x), \qquad P_0 = 1, \quad P_1 = 2x,$$

and are orthogonal with respect to

$$W(x) = |\Gamma(\tfrac{1}{4} + ix)|^2 = \Gamma(\tfrac{1}{4} + ix)\,\Gamma(\tfrac{1}{4} - ix), \qquad \int_{\mathbb{R}} P_m P_k\, W\, dx = h_k\,\delta_{mk}.$$

Write $\widehat{P}_k = P_k/\sqrt{h_k}$ for the orthonormal polynomials. The *Meixner–Pollaczek functions* are

$$\Phi_m(x) = \widehat{P}_m(x)\,\sqrt{W(x)},$$

an orthonormal system in $L^2(\mathbb{R}, dx)$ — the analogue of the Hermite functions $\psi_m = c_m H_m e^{-t^2/2}$.

Remark 1.1 (Bump–Ng)**.** The Mellin transform of the Hermite function factors as $M\psi_{2n}(s) = c_{2n}\,2^{s/2-1}\,\Gamma(s/2)\,Q_n(s)$ with Q_n a polynomial. On the critical line $s = \frac{1}{2} + 2ix$ this becomes, up to a phase, $\Gamma(\frac{1}{4}+ix)\,P_n^{(1/4)}(x)$: the Meixner–Pollaczek functions *are* the Mellin transforms of the (even) Hermite functions. Odd Hermite functions give $\lambda = 3/4$. Since $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ carries the same Γ-factor, expansions of ξ on the critical line live naturally in $L^2(W)$.

2 The Matching Matrix

The best $L^2(W)$-approximation of Φ_m by polynomials of degree at most N is $\sum_{k\le N} b_{mk}\widehat{P}_k$ with

$$b_{mk} = \langle \Phi_m, \widehat{P}_k\rangle_W = \int_{\mathbb{R}} \widehat{P}_m(x)\,\widehat{P}_k(x)\,|\Gamma(\tfrac{1}{4} + ix)|^3\,dx, \tag{1}$$

the Γ-cubed analogue of the Hermite integrals $\int H_m H_k\, e^{-3t^2/2}\,dt$.

Proposition 2.1. *$b_{mk} = b_{km}$; $b_{mk} = 0$ when $m + k$ is odd; and numerically* $\operatorname{sign}(b_{mk}) = (-1)^{(m+k)/2}$ *for all computed entries $m, k \le 11$ — the same checkerboard as the Hermite matrix.*

The symmetry and parity are immediate from (1) (the weight is even and $P_n(-x) = (-1)^n P_n(x)$).

Tables 1 and 2 give the two parity blocks, computed by `mpmath` quadrature at 40 digits (`mp_matching_table.py`; cache `mp_matching_table.pkl`).

Table 1: b_{mk}, even block.

$m\backslash k$	0	2	4	6	8	10
0	2.6560	−0.6180	0.3542	−0.2517	0.1970	−0.1629
2	−0.6180	0.5739	−0.3590	0.2548	−0.1961	0.1592
4	0.3542	−0.3590	0.2981	−0.2341	0.1882	−0.1558
6	−0.2517	0.2548	−0.2341	0.2023	−0.1720	0.1471
8	0.1970	−0.1961	0.1882	−0.1720	0.1533	−0.1356
10	−0.1629	0.1592	−0.1558	0.1471	−0.1356	0.1235

Table 2: b_{mk}, odd block.

$m\backslash k$	1	3	5	7	9	11
1	1.1422	−0.5190	0.3311	−0.2430	0.1923	−0.1594
3	−0.5190	0.4116	−0.2969	0.2264	−0.1814	0.1509
5	0.3311	−0.2969	0.2487	−0.2042	0.1699	−0.1442
7	−0.2430	0.2264	−0.2042	0.1787	−0.1551	0.1352
9	0.1923	−0.1814	0.1699	−0.1551	0.1395	−0.1249
11	−0.1594	0.1509	−0.1442	0.1352	−0.1249	0.1144

3 What Does Not Survive from the Hermite Case

3.1 No algebraic structure

In the Hermite case, the integrals $\int H_m H_k\, e^{-3t^2/2}$ are integer multiples of $\sqrt{2\pi/3} \cdot 3^{-(m+k)/2}$, with a signed matching interpretation (cross-pairs weight $+4$, self-pairs -2). Here, PSLQ (`mpmath.identify`) finds no algebraic relation for any ratio J_{mk}/J_{00} of the raw integrals $J_{mk} = \int P_m P_k\, W^{3/2}$, and

$$J_{00} = \int_{\mathbb{R}} |\Gamma(\tfrac{1}{4} + ix)|^3\, dx = 20.91540486643\ldots$$

matches no closed form in π, $\Gamma(1/4)$, $\Gamma(1/4)$'s companion $\Gamma(3/4)$, or their square roots. The Hermite miracle rests on the self-reproducing Gaussian generating function e^{2tz-z^2}; the weight $|\Gamma|^3$ is a *half-integer* power of the Barnes kernel $\Gamma(\frac{1}{4}+ix)\Gamma(\frac{1}{4}-ix)$, outside the reach of Barnes' lemmas.

3.2 Algebraic, not geometric, decay

The Hermite matching coefficients decay geometrically, $a_{0k} \sim 3^{-k/2}$. The Meixner–Pollaczek coefficients do not:

Observation 3.1. *Out to $k = 40$ (row $m = 0$, 60-digit quadrature), the local log-log slope of $|b_{0k}|$ drifts slowly downward through -0.84, consistent with*

$$b_{0k} \sim C\, k^{-\alpha}, \qquad \alpha \approx \tfrac{3}{4},$$

possibly with logarithmic corrections.

k	$\ln \lvert b_{0k} \rvert$	local slope in $\ln k$
10	-1.815	-0.86
20	-2.405	-0.85
30	-2.747	-0.85
40	-2.989	-0.84

The obstruction is visible in the complex plane. The Gaussian weight is entire, so $\sqrt{e^{-t^2}}$ is entire and the projection error decays geometrically. But $W(x) = \Gamma(\frac{1}{4} + ix)\Gamma(\frac{1}{4} - ix)$ has poles at $x = \pm i/4$, so $\sqrt{W}$ (and $W^{3/2}$) has branch points at distance $1/4$ from the real axis. A singularity at finite distance forbids geometric convergence of a polynomial expansion; the branch order governs the algebraic exponent α. Pinning down α exactly (natural candidates: $3/4$, $5/6$) is open.

3.3 The matrix is not band-dominated

At $m, k \approx 10$ all entries still have magnitude ≈ 0.12–0.15: the matrix decays slowly *toward the diagonal from both sides*, and each Φ_m genuinely requires all polynomial degrees.

4 Zeros of the Approximants

Let $\Phi_m^{(N)} = \sum_{k \le N} b_{mk} \widehat{P}_k$ denote the degree-N approximant. In the Hermite case the approximants develop a band of *complex* spurious zeros beyond the onset $|t| \sim \sqrt{N}$, arranged in arcs. The Meixner–Pollaczek picture (Figure 1; `mp_approximant_zeros.py`) differs in both respects:

Observation 4.1. *For $m = 0, 4$ and $N \le 40$, the zeros of $\Phi_m^{(N)}$ are all real (first exception: four complex zeros at $m = 0$, $N = 40$, with $|\mathrm{Im}| \le 0.1$), and they spread linearly, out to $|x| \approx 35$ at $N = 40$ — consistent with the balance-law prediction $W(x^*) \approx b_N^2$ for an exponentially decaying weight.*

The realness is the equioscillation phenomenon: the best $L^2(W)$-polynomial approximation to the positive function $\sqrt{W}$ (times $\widehat{P}_m$) oscillates *around* its slowly decaying exponential tail, crossing it on the real axis, rather than peeling off into the complex plane as in the super-exponentially decaying Gaussian case.

The price is paid elsewhere: at $m = 4$, $N = 40$ the four *genuine* zeros of P_4 (± 0.2967, ± 1.6315) are *not* captured — the nearest approximant zeros sit at ± 0.3622 and ± 1.3616, errors of order 10^{-1} after 36 extra degrees of freedom. In the Hermite case the genuine zeros of H_4 were reproduced to many digits at the same depth. This is the algebraic decay of Observation 3.1 made visible: the truncation error at $N = 40$ is still $\sim 10^{-1}$, and the approximant spends its zeros equioscillating around the tail instead of resolving the head. The approximant zeros also do not coincide with the zeros of P_{40} (Figure 1, right): near the origin the two sets are visibly distinct, so $\Phi_4^{(40)}$ is not simply shadowing its top basis element.

5 Discussion

The practical consequence for the zeta programme: any function carrying the archimedean factor $\Gamma(s/2)$ — including ξ itself on the critical line — admits only *algebraic-rate* approximation by Meixner–Pollaczek polynomials. This quantifies the earlier numerical finding that the degree-60 polynomial approximation of Ξ captures two Riemann zeros while the Hermite-function *sections* (which carry the weight-root along and suffer no branch-point mismatch) capture five or more. In the trade between polynomial bases and function bases, the Γ-factor's pole at $s = 0$ — equivalently the branch point of $\sqrt{W}$ at $x = i/4$ — is the entire cost.

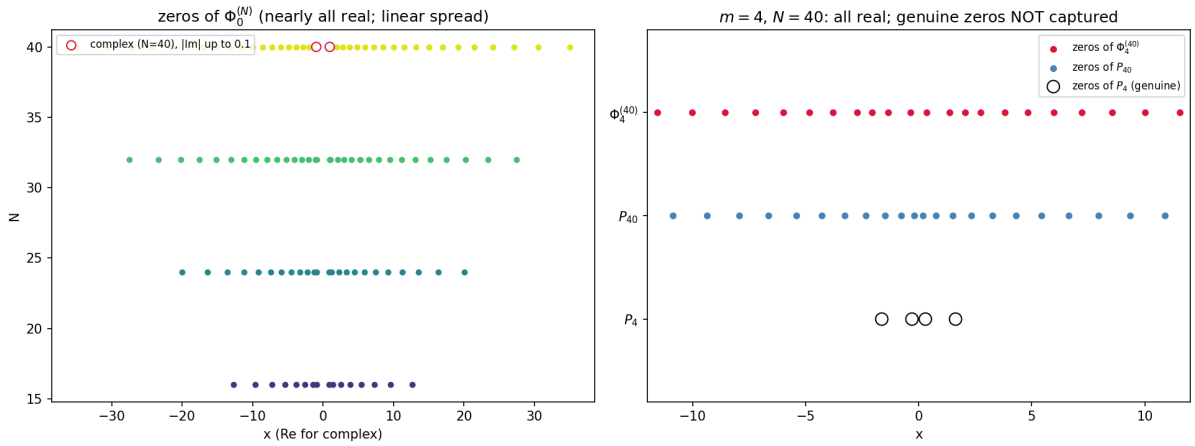


Figure 1: Left: zeros of $\Phi_0^{(N)}$ for $N = 16, 24, 32, 40$ (all real except four at $N = 40$, open circles), spreading linearly with N. Right: the 40 real zeros of $\Phi_4^{(40)}$ (top row) against the zeros of P_{40} (middle) and the genuine zeros of P_4 (bottom), restricted to $|x| \le 12$.

Open questions: the exact decay exponent α; the large-k factorisation $b_{mk} \sim f(m)\, g(k)$ suggested by the near-rank-one look of the far off-diagonal entries; and whether the $\lambda = 1/4$ and $\lambda = 3/4$ blocks (even and odd Hermite functions under Mellin) interact through the joint weight, which the reflection formula evaluates in closed form: $|\Gamma(\frac{1}{4} + ix)|^2\, |\Gamma(\frac{3}{4} + ix)|^2 = 2\pi^2 / \cosh(2\pi x)$.